**Edwin Arkel Rios** Personal Page
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Architect of efficient deep learning systems, specializing in model compression and HW-SW co-design.

## Education

**National Yang Ming Chiao Tung University** Hsinchu, Taiwan
*PhD in Electrical Engineering and Computer Science, Cumulative GPA: 4.24/4.30* *July 2019 - Present*

**National Cheng Kung University** Tainan, Taiwan
*Bachelor of Science in Energy Engineering, Cumulative GPA: 3.76/4.30* *Sept. 2015 - June 2019*

## Experience

**Tranxform AI, Generative AI R&D Intern** *December 2025 - September 2026*

- Developed a data-free post-training quantization (PTQ) algorithm to quantize large audio language models (LALMs) to 1.58-bit precision, enabling up to 20x memory reduction (e.g., 28GB → ∼1.4GB for 7B models).
- Built a quantized automatic speech recognition (ASR) framework in Hugging Face transformers (32 quantization configs, 26+ models, 100+ datasets, 50+ languages) used in 6 papers under review.

**NYCU PCS Lab & NTHU MIS Lab, Research Assistant** *July 2019 - Present*

- Developed methods for knowledge distillation, pruning, and efficient training of vision models, leading to 7 first-author papers in international venues: ICASSP, ISCAS, ECCV Workshop, and CVPR Workshop.
- Led and mentored 20+ students, resulting in 10 publications and 3 awards at the school and national levels.

## Selected Research Projects

**Low-Bit Quantization of Large Audio Language Models** *2025 - 2026*

- Benchmarked 18 ASR and audio language models across 25 languages at 3 to 8 bits: accuracy loss tracks model size, not per-language training data; distilled 1.58-bit decoders with 10x compression.

**Input Pruning for Acceleration of Transformers via HW-SW Co-Design** *2023 - Present*

- Designed a module to support pruning of up to 90% of inputs to reduce FLOPs while maintaining accuracy.
- Proposed a methodology to select the optimal settings for token reduction in an FPGA accelerator, increasing FPS by up to 20x and achieving 7x higher energy efficiency compared to SOTA accelerators.

**Parameter-Efficient Fine-Tuning (PEFT) of Foundation Models** *2023 - Present*

- Developed a lightweight adapter to adapt large models with limited resources, improving the accuracy by 5.1% over baselines while requiring 123x fewer trainable parameters.

**Efficient Discriminative Feature Selection for Fine-Grained Image Recognition (FGIR)** *2021 - 2024*

- Engineered a novel mechanism for selecting high importance regions for fine-grained classification, requiring up to $10^7$ times fewer FLOPs while improving classification performance by 10.15%.

## Skills

**Languages:** Spanish (native), English (fluent), Mandarin Chinese (proficient)

**Programming Languages:** Python, MATLAB, C++, Bash

**Libraries:** PyTorch, Hugging Face transformers, HQQ, bitsandbytes, scikit-learn, OpenCV, NumPy, pandas

**Tools:** Git, Docker, Linux, Weights & Biases, Vitis HLS for Xilinx FPGAs, Jupyter Notebooks, LaTeX, HTML

## Awards and Certifications

**Merit Paper Award**, 39th IPPR Conference on CV, Graphics and Image Processing (CVGIP) *2026*

**Excellent Paper Award**, 38th IPPR Conference on CV, Graphics and Image Processing (CVGIP) *2025*

**Associate Fellow (AFHEA)** of AdvanceHE, UK Professional Standards Framework in Higher Education *2025*

**NCTU Academic Achievement Award** during R.O.C. Academic Semester 108-1 *2020*

**NCKU Outstanding Student** during R.O.C. Academic Year 105 and 106 *2018*

## Selected Research Works

7 first-author publications (ICASSP, ISCAS, ECCV Workshop, CVPR Workshop); 1 ICLR talk; 6 under review.

How to Choose Your Teacher for Fine Grained Image Recognition *CVPR W 26*

Cross-Layer Cache Aggregation for Token Reduction in Ultra-Fine-Grained Image Recognition *ICASSP 25*